Say Hello To The World of Git

All you need is a lil' bit of "commit"ment!

Introduction to Git

Git, to put it simply, is a version control system that was developed by Linus Torvalds... the same GOAT who developed Linux.

A version control system is used to monitor changes in files and folders during collaborative work.

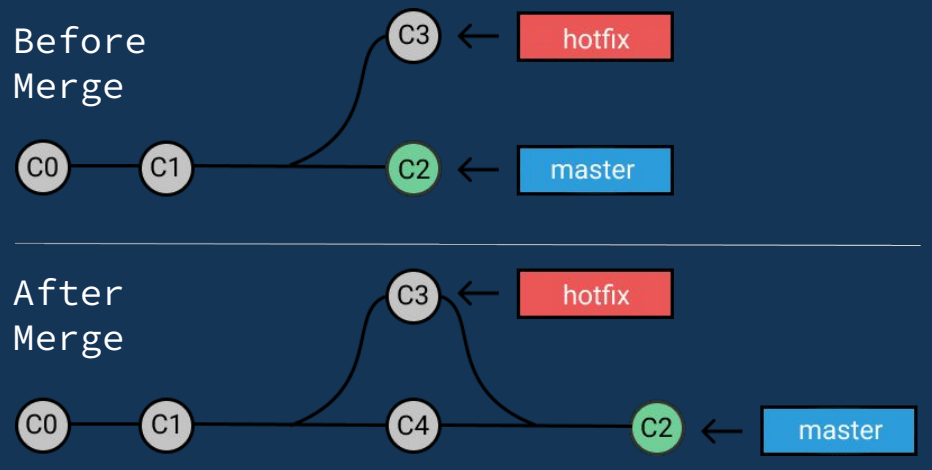
```
''' Fun fact:  Git is a British slang
    meaning an 'unpleasant person'.
    Torvalds named git after himself
    saying "I'm an egotistical bastard
    and I name all my projects after myself.' '''
```

Yes this is the guy who created Linux and Git as personal projects

Common terms related to Git and GitHub

Repository	:- Folder containing the project, generally called a repo
Branch	:- Simultaneous instances of your repository in different states
Pull Request (PR)	:- Request to moderator of the repo to merge a branch
Clone	:- Make a local copy of a repository from an online source
Add or Stage	:- Add files to staging area
Commit	:- Mark a change point (like saving a game)
Push	:- Literally pushing files to online repository

HOW GIT WORKS



Source:
https://dev.to/neshaz/how-to-use-git-merge-the-correctway-25pd

Git supports non-linear development.

Collaborators make use of 'branches' to make their own changes to files and folders in the 'main' branch.

These changes can then be merged to the main branch after review.

As git has gone under major development over the years, now it supports a wide variety of features such as distributed development, cryptographic authentication of history, and many others.

Some useful git commands

git init [name]	initializes a new git repository [name]
git add [file name. extension]	adds a file to the staging area (think of the stage like an auction. The item has been presented but not sold yet)
git branch [branch name]	creates a new branch
git branch -d [branch name]	deletes the mentioned branch
git checkout -b [branch name]	creates a new branch and switches to it
git status	displays the status of the git repository

Useful Tip : using ‘git add *’ adds all the changed files to the staging area

Some more useful commands

git commit -m "[commit message]"	commits changes with a message (now the item has been sold)
git rm -r [file name. extension]	removes file or folder
git merge [source branch] [target branch]	merges source to target
git push [remote name] [branch name]	pushes local changes to branch of remote repo
git pull [remote name] [branch name]	pulls all changes from branch of a remote repo

What is GitHub?

- GitHub is a cloud-based service for development and version control using Git.

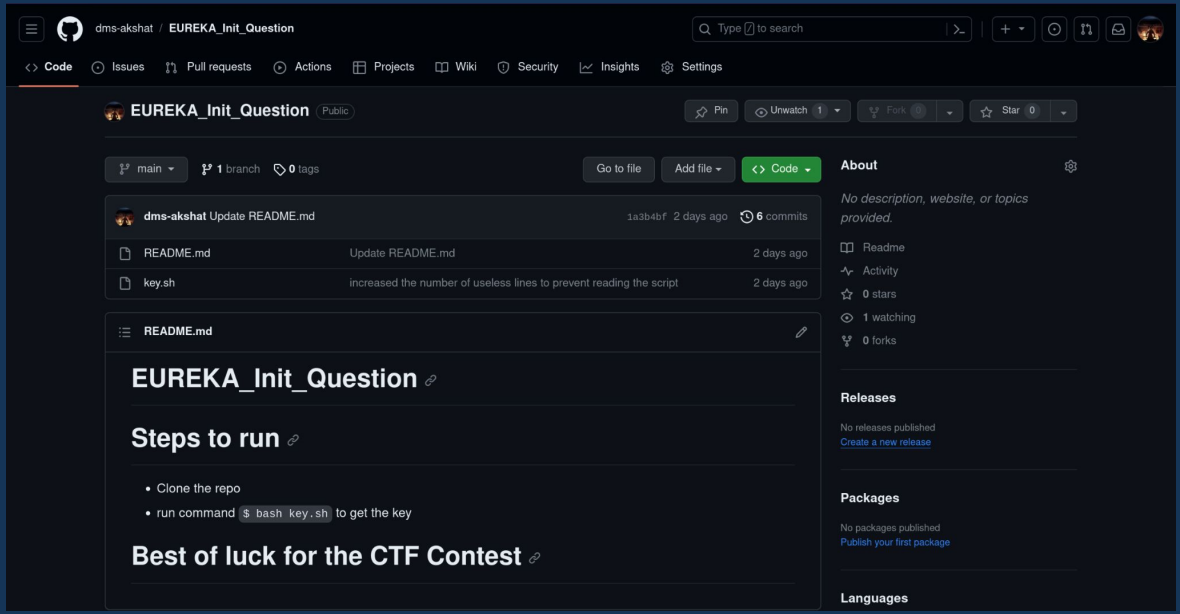
- GitHub allows users to store and manage their code. It provides the version control features of git along with access control to repos, bug tracking, task management, and much more.

- GitHub has been widely used to collaborate on open-source projects and is the world's largest source code hosting site as of June 2023.

WHAT IS GITHUB?

Git can be used to modify repos on GitHub. Git is used mainly for version control on your local machine before making changes on GitHub.

GitHub has a nice UI for making pull requests, logs, and other features.

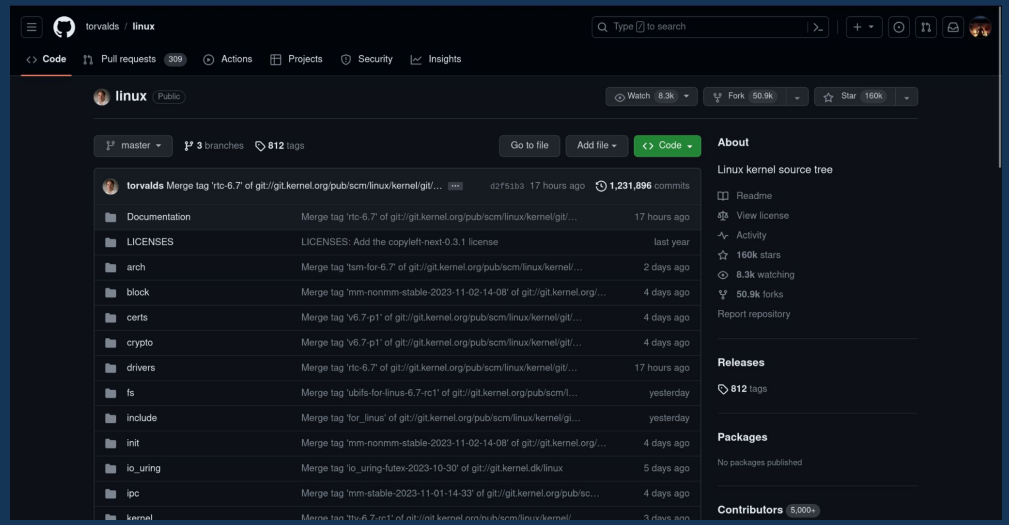


Difference between Git and GitHub

hub

WHY USE GITHUB??

- Best way to share your code with the world
- Easy to collaborate with team members
- Easy to manage bugs and issues in extremely large code bases
- Cloud based so your code can be accessed from any system
- Acts like a off site backup for your codebase



Hands on with Git and Github

Everyone, keep your terminals ready

!!It's SHOWTIME!!

Step 1 - Clone the given repo -

```
$ git clone https://github.com/dms-akshat/EUREKA_Init_Question
```

Step 2 - Open the cloned directory in your system -

```
$ cd EUREKA_Init_Question
```

Step 3 - Run the BASH script present in the directory -

```
$ bash key.sh
```

Congratulations, you just got your first key for CTF
Keep it up
Best of Luck for the Event

Have Fun :)